

## Nathan Adrian Saldanha

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[linkedin.com/in/nathan-adrian](https://www.linkedin.com/in/nathan-adrian) • [github.com/HyperToken9](https://github.com/HyperToken9) • [Figma Portfolio](#)

### Education

#### Manipal Institute of Technology

Bachelor of Technology (B.Tech) in **Computer Science Engineering**  
Minor in Computational Intelligence. **GPA: 8.73**

Manipal, Karnataka

Graduated July 2025

### Professional Experience

#### ZS Associates

##### Business Technology Associate

Gurgaon, Haryana

June 2025 - Present

- **Full-Stack Cloud Applications:** Developed and maintained custom web applications utilizing **React** and **FastAPI**.
- **Search Optimization:** Architected an LLM-powered query expansion on **cloud managed services** (AWS Bedrock, OpenSearch), boosting **search relevance by 300%** while surfacing implicit scientific data inaccessible via standard metadata.
- **Data Engineering:** Deployed **cloud data pipelines** (PySpark, Databricks) to process 7GB+ datasets, implementing **BI-as-Code** to streamline visualization workflows across multiple data products.

### Technical Experience

#### Project MANAS

*Artificial Intelligence Member - Python, C++, Robot Operating System (ROS), OpenCV*

Manipal, Karnataka

June 2022 - April 2024

- Integrated a planning and navigation system for an autonomous drone that **placed second at the SUAS 2023 competition**. [\[demo\]](#)
- Created a computer vision pipeline for an Unmanned Ground Vehicle to localize objects with **<10 cm accuracy**, leading to a **third-place finish at IGVC 2023**. [\[demo\]](#)

#### Kasturba Medical College

*Application Developer - NextJS, Python, Scikit-learn* [\[source\]](#), [showcase](#)

Manipal, Karnataka

May 2024 - December 2024

- Optimized a **logistic regression model** to predict surgical interventions during childbirth. Implemented stratified K-fold cross-validation and feature normalization, achieving an **AUC of 0.97** against a baseline of 0.853.
- Hosted a NextJS web application on Vercel, providing an interface for clinical professionals to interact with and apply predictive models in real-world settings.

### Projects

**Autonomous Drone Control via RL** - Python, Stable Baseline 3, PyFlyt [\[demo\]](#)

March 2026

- Trained a drone controller for high-speed flight using **Proximal Policy Optimization (PPO)** in a custom Gymnasium environment.
- Defined an ego-centric state space and tuned reward functions for PRYT (Pitch, Roll, Yaw, Thrust) control.

**Text-to-Handwriting** - Python, PIL, Angular, Flask [\[source\]](#), [showcase](#)

October 2021

- Built a web app using **Angular** for a responsive frontend and **Flask** for a high-performance image processing backend.
- Defined a layout algorithm to simulate natural handwriting by applying distortions to character size and placement.
- Leveraged image augmentation to transform text into realistic handwritten documents using a custom character dataset.

**C-ROS: A Middleware Library in C** - C, CMake [\[github\]](#)

November 2023

- Designed an inter-process communication (IPC) library to replicate the core functionality of ROS.
- Implemented a thread-safe publisher-subscriber model using a custom socket-based message broker and mutex locks for synchronization.
- Optimized data transmission efficiency to achieve **<200  $\mu$ s message latency** across distributed nodes

**AutoShorts: Reddit-to-YouTube Video Generator** - Python, PRAW, MoviePy, FFmpeg [\[demo\]](#)

May 2022

- Developed an end-to-end content pipeline that scraped high-engagement Reddit threads and transformed them into vertical-format videos.
- Programmatically synthesized voiceovers, dynamic captions, and background gameplay footage (Minecraft/GTA) using MoviePy and FFmpeg wrappers.
- Scaled a YouTube channel to **166K+ views and 1,200+ watch hours** entirely through automated content creation.

### Skills

**Languages:** Python, C/C++, Dart, JavaScript | **Frameworks:** Flutter, React, Tailwind CSS, Angular

**Cloud & AI:** AWS (Lambda, Batch, Bedrock), OpenSearch, Databricks, OpenCV | **Databases:** PostgreSQL, DuckDB